

1. Operationalizing Autonomous Commerce for Retail

Farooq Adam

A roadmap for moving retail operations from human-driven to autonomous.

1.1 Executive Summary

Retail is an inventory risk business. 85-97% of every rupee a retailer earns goes toward acquiring, storing, moving, funding, and selling inventory. The remaining 3-15% is profit - the reward for getting the bet right. This is not a metaphor. It is the arithmetic of every annual report from Inditex to DMart to Reliance Retail. Revenue is selling inventory. COGS is buying inventory. Rent is storing inventory. Employees manage inventory. Everything else is a derivative.

Today, humans place this bet. They forecast demand, select suppliers, set prices, plan promotions, allocate stock, and decide markdowns. AI has entered the conversation - but in most organisations, it is still a surface-level addition. The spreadsheet is better-formatted. The PowerPoint is faster. The forecast has one more input. The bet itself has not changed.

Autonomous Commerce changes this. Not incrementally. Structurally. It is a framework for deciding, per workflow, the right level of autonomy - what should be fully autonomous, what should be AI-assisted, and what should remain human. The question is not "how do we use AI in retail?" The question is "at what level of autonomy should each workflow operate, and what must be true to move it to the next level?"

The approach covers both technology and organisation. On the technology side, the platform connects AI to every workflow and data stream - demand signals, inventory positions, supplier networks, pricing engines, and fulfilment nodes become a single addressable surface. The data catalog (Paper 2) documents the 67 activities, 120 SOPs, and 35 data sources that power these workflows. On the organisation side, each function must make a deliberate choice about its target autonomy level - and the people freed by that transition are redirected into AI-native roles that did not exist before. This paper introduces the framework and maps it across the retail value chain for Retail.

1.2 The Risk Gates: Guardrails

The transition between autonomy levels is not a technology upgrade. It is a risk reduction proof.

Every retail workflow sits at some level of autonomy today. Moving it to the next level requires answering one question: does this move demonstrably reduce risk? Not "can AI do this?" - that is a technology question. Not "will this save headcount?" - that is an efficiency question. The question is whether giving the machine more authority over this workflow produces measurably better outcomes with measurably lower risk. If you cannot prove it, you are not ready. If you can prove it but the organisation will not act on the proof, the problem is not technology. It is governance.

Risk manifests differently across the value chain. **Inventory holding risk** governs Plan. **Procurement risk** governs Buy. **Supply chain risk** governs Move. **Revenue risk** governs Sell. Four domains, four categories of risk, one standard: here is the data, here is the A/B test, here is the measurable outcome in rupees. This rule governs everything that follows. A detailed note on Risks and Guardrails will be published in the next two weeks.

1.3 The Autonomy Ladder

Six levels of autonomy, adapted from the Autonomous Networks taxonomy standardised by TM Forum for telecommunications. The domains change - from network fault management to demand forecasting, from radio access to supply chain - but the architecture of autonomy is universal. Each level defines who decides, who executes, and who bears risk.

L0 - Manual Commerce. Humans execute all tasks. Systems are record-keeping tools. Monitoring and data capture only. All risk borne by human judgement and experience.

L1 - Assisted Commerce. Humans make all decisions. AI performs specific, repetitive subtasks on pre-configured rules - report generation, data formatting, alert triggering. Risk unchanged. Execution speed improved.

L2 - Augmented Commerce. Humans own decisions but operate within AI-generated recommendations. Closed-loop execution for narrow, well-defined units. Risk reduced in narrow domains with proven rule sets.

L3 - Intelligent Commerce. Humans set policies and handle exceptions. The system senses real-time changes and self-optimises within defined boundaries - adjusting to demand shifts, competitive moves, and supply disruptions. Risk actively managed by the system within guardrails.

L4 - Agentic Commerce. Humans set strategic intent and review outcomes. The system makes cross-domain decisions via predictive analysis and continuous learning. Coordinates across Plan, Buy, Move, and Sell simultaneously. Risk predicted and pre-empted.

L5 - Autonomous Commerce. Humans define purpose, values, and constraints. The system operates the full lifecycle via cognitive self-adaptation - self-healing, self-optimising, continuously learning. Risk is structurally minimised.

The jump from L1 to L2 is a tool upgrade - the same people with better instruments. The jump from L2 to L3 is a skill upgrade - the same people, doing the same jobs, faster and with higher leverage. The jump from L3 to L4 is an identity shift. The roles themselves dissolve, and the organisation becomes a fundamentally different kind of entity.

Not every workflow should reach L5. This is a design choice, not a failure of ambition. Workflows that are high-frequency, data-rich, and reversible - demand forecasting, pricing, replenishment, allocation - move toward L4-L5. Workflows that are low-frequency, judgement-intensive, and irreversible stay deliberately at lower levels. Strategy and vision (what categories to enter, what the brand stands for) stay at L1-L2: AI-informed, human-decided. Brand and creative direction stays at L3: AI-generated, human-curated. Strategic supplier relationships stay at L3-L4: AI-managed operationally, human-led strategically. Compliance and ethical judgements stay at L2-L3: AI-monitored, human-accountable. The right answer is not a uniform target. It is a portfolio of autonomy levels - chosen deliberately, per workflow, governed by provable risk reduction.

1.4 Autonomy Framework Across the Retail Value Chain

The matrix maps retail workflows against the six autonomy levels. Read it left to right to see how a single workflow evolves from fully manual to fully autonomous. Read it top to bottom within a value chain stage to see the breadth of autonomy decisions an organisation must make simultaneously. The expanded matrix - with full capability descriptions per cell - is rendered as a separate diagram artefact. The workflows below map directly to the 67 activities, 120 SOPs, and 35 data sources documented in the data catalog (Paper 2).

#	Workflow	L0 Manual	L1 Assisted	L2 Augmented	L3 Intelligent	L4 Agentic	L5 Autonomous
PLAN							
1.1	Demand Forecasting	Spreadsheets and instinct	AI summarises trends	ML forecasts, human reviews	Continuous signal engine	Forecast triggers actions	Demand is sensed, not forecast
1.2	OTB Budgeting	Excel, quarterly cycle	AI formats the deck	AI proposes, planner reviews	Auto-reallocating constraint	Cross-format OTB, human sets intent	Real-time financial envelope
1.3	Range Architecture	Hand-built option plans	AI generates trend reports	AI-generated options, buyer chooses	System generates full range plans	Human scans and asks "what if?"	Signal to order in days
1.4	Space Planning	Fixed planograms, regional clusters	AI suggests layouts	Weekly GMROF optimisation	Continuously optimised per store	Daily VM briefs per store	Every store is a market of one
1.5	Assortment Planning	Periodic, cluster-level	AI surfaces bestsellers	Store-cluster, AI recommends	Store-level, demand-driven	Cross-format cannibalisation detection	Unified demand sensing
BUY							
2.1	Sourcing	Personal network, trust-based	AI benchmarks prices	System scores and recommends	Real-time supplier graph	System selects, negotiates, orders	Autonomous within guardrails
2.2	Costing	Instinct-based negotiation	System tracks indices	Target cost per SKU pre-negotiation	Predicted landed cost pre-spec	Predictive, real-time margin	System-to-system pricing
2.3	Product Development	Multi-month physical cycle	AI generates concepts	Digital prototyping, partial	Majority digital approval	Spec-to-order in days	Generative design, direct-to-production
2.4	Order Placement	Bulk order, over-ordering	Forecasts inform quantities	Partial order, reserve capacity	Rolling orders, weekly adjustment	Trend-to-order, human reviews	Continuous, no order window
2.5	Vendor Management	Quarterly reviews, spreadsheets	Auto-generated scorecards	System flags at-risk vendors	Predictive risk identification	System manages operationally	Self-governing ecosystem
MOVE							

#	Workflow	L0 Manual	L1 Assisted	L2 Augmented	L3 Intelligent	L4 Agentic	L5 Autonomous
3.1	Production Tracking	Email and phone calls	AI monitors milestones	Digitised milestone tracking	Predictive completion modelling	Delay predicted, mitigation auto-identified	Orchestrated, not tracked
3.2	Quality Assurance	Warehouse inspection, high defects	Risk-scored inspection triggers	Selective QC replaces blanket	Pre-production confidence scoring	Quality built in, not inspected	Zero-defect, system-to-system
3.3	Allocation	Once per cycle, A/B/C grading	AI considers store-level demand	SKU-store level, one-time	Continuous, demand-signal driven	Chase and replenish are one action	Inventory is a network
3.4	Replenishment	Manual, weekly, store calls	Auto-trigger, static thresholds	Smart slow-seller / fast-seller	Dynamic, real-time thresholds	Self-healing supply chain	Nervous system, autonomous
3.5	Logistics	Carrier negotiation, 3PL routing	Optimised routing, TAT tracked	Ship-from-store, intelligent routing	Cross-network path optimisation	Auto-rerouting on disruption	Invisible, self-optimising
SELL							
4.1	Store Operations	Weekly visits, monthly reviews	Real-time dashboards	Same-day KPI deviations	Every store a market of one	Daily briefs, guardrails not instructions	Demand-sensing node
4.2	Visual Merchandising	Brand guidelines, periodic	AI recommends placement	Data-driven display rules	Daily recommendation per store	Autonomous daily planograms	Continuous, every fixture optimised
4.3	Pricing and Markdowns	Calendar-based clearance	SKU-level markdown recommendations	Dynamic, product-driven timing	Continuous, store-specific	Every SKU, every store, every day	Price is a real-time function
4.4	Promotions	Central, uniform across stores	AI recommends where to promote	Cluster-level promotion variation	Opportunity-detected, instant approval	Continuous micro-campaigns	No campaigns, individual conversations
4.5	Omnichannel	Online and offline separate	Unified inventory	Intelligent fulfilment routing	Single pool, channel-agnostic	Cross-format lifecycle management	Shopping disappears as effort
4.6	Marketing	Calendar-based campaigns, broadcast	AI writes copy, resizes images	Data-driven segmentation, automated bidding	Micro-campaigns triggered by real-time signals	Autonomous audience, creative, deploy, measure	Continuous individual conversations
ASSET							
5.1	Network Planning	Manual site scanning, spreadsheet feasibility	AI-assisted catchment analysis	System scores and ranks potential sites	Predictive demand modelling per catchment	System recommends network expansion plan	Self-optimising network topology
5.2	New Store Opening	Manual checklists, sequential approvals	Digitised readiness tracking	System monitors milestones, flags delays	Predictive fit-out scheduling	Autonomous pre-opening orchestration	Self-configuring store launch

#	Workflow	L0 Manual	L1 Assisted	L2 Augmented	L3 Intelligent	L4 Agentic	L5 Autonomous
5.3	Maintenance and Upkeep	Reactive, ticket-based	Automated work order generation	Predictive maintenance alerts	IoT-driven condition monitoring	Self-healing facility management	Autonomous asset lifecycle
5.4	Security and Loss Prevention	Manual audits, guard-based	CCTV with AI anomaly detection	Automated shrinkage monitoring	Predictive loss modelling per store	Autonomous incident response	Self-governing security network
FINANCE							
6.1	Revenue and Margin Tracking	Monthly manual consolidation	Automated dashboard generation	Real-time margin visibility	Predictive margin alerts and recommendations	System manages margin levers autonomously	Self-optimising P&L
6.2	Accounts Payable	Manual invoice matching	Automated three-way match	Exception-based processing	Predictive payment scheduling	Autonomous vendor payment execution	Self-governing payables ecosystem
6.3	Tax and Compliance	Manual filing and reconciliation	Automated GST/TDS computation	System flags compliance gaps	Predictive audit preparation	Autonomous regulatory filing	Self-monitoring compliance
HR							
7.1	Workforce Planning	Manual headcount planning	AI-informed staffing models	System recommends roster optimisation	Demand-driven dynamic scheduling	Autonomous workforce allocation	Self-balancing workforce network
7.2	Training and Development	Calendar-based training programmes	AI recommends learning paths	Personalised skill gap analysis	Adaptive learning, completion tracking	AI-curated development journeys	Continuous capability building
7.3	Performance Management	Annual review cycles	Continuous KPI dashboards	System surfaces performance patterns	Predictive attrition and engagement signals	Autonomous goal-setting and feedback	Self-governing talent ecosystem

The matrix is a workflow tool. Its deepest consequence is organisational. Once you set the target level per workflow, the org chart follows - because every role is defined by the workflows it touches. The move from L2 to L3 does not change the org chart. The planner is still a planner, the buyer is still a buyer - faster, better-informed, operating at higher leverage. The move from L3 to L4 changes what the organisation is. Planning teams compress from dozens to a handful of strategic minds who set intent and handle exceptions. Buying teams become acquisition strategists who manage supplier relationships and review AI-generated decisions. Supply chain teams become strategic operators who handle the disruptions that require human judgement. The function that creates content operates at L4 - AI-led, human-steered. The function that sets vision operates at L1 - AI-informed, human-decided. These are not the same conversation, and they cannot be governed by the same policy.

Not everyone will step up. The honest reality is that L4 demands a capacity and ambition that not every person on every team possesses. The commitment is that the organisation will redirect, not displace - into work that did not exist before. The condition is equally clear: its people must become AI-native.

1.5 Autonomous Commerce at Retail (Recommendations for Next 90 Days)

The heat map on the following page shows the recommended autonomy level for each workflow achievable in the next 90 days - given current data infrastructure, current technology, and current organisational readiness. The 90-day roadmap focuses on Plan, Buy, Move, and Sell.

Across the 21 core retail workflows, L2 (Augmented) is the baseline - every workflow reaches at least this level. Of these, 43% move to L3 (Intelligent) where the system optimises within guardrails and humans handle exceptions. 29% reach L4 (Agentic) where the system makes cross-domain decisions and humans set intent. The remaining 24% operate at L2 where AI recommends and humans decide.

The subsequent papers in this series lay out the implementation of this autonomy framework for Fashion and Lifestyle:

- Paper 2: State of Data Catalog
- Paper 3: Command Centre
- Paper 4: Details of AI Agents

Workflow	L0 Manual	L1 Assisted	L2 Augmented	L3 Intelligent	L4 Agentic	L5 Autonomous
PLAN						
1.1 Demand Forecasting						
1.2 OTB Budgeting						
1.3 Range Architecture						
1.4 Space Planning						
1.5 Assortment Planning						
BUY						
2.1 Sourcing						
2.2 Costing						
2.3 Product Development						
2.4 Order Placement						
2.5 Vendor Management						
MOVE						
3.1 Production Tracking						
3.2 Quality Assurance						
3.3 Allocation						
3.4 Replenishment						
3.5 Logistics						
SELL						
4.1 Store Operations						
4.2 Visual Merchandising						
4.3 Pricing & Markdowns						
4.4 Promotions						
4.5 Omnichannel						
4.6 Marketing						
ASSET						
5.1 Network Planning						
5.2 New Store Opening						
5.3 Maintenance & Upkeep						
5.4 Security & Loss Prevention						
FINANCE						
6.1 Revenue & Margin Tracking						
6.2 Accounts Payable						
6.3 Tax & Compliance						
HR						
7.1 Workforce Planning						
7.2 Training & Development						
7.3 Performance Management						

2. State of Data Catalog for F&L Retail

2.1 Data Catalog & Workflow

This paper establishes the **data foundation** - what data exists, how it is classified, how it is layered - and then maps it directly to the **end-to-end F&L value chain**, showing exactly which data powers which workflow, who is accountable, and how performance is measured.

The data catalog serves two distinct layers, aligned to the ARTS (Association for Retail Technology Standards) framework: the **Operational Data Model** powers the systems that run the business in real time - current inventory positions, live order status, active pricing - optimised for transactional correctness and current state. The **Data Warehouse Model**, derived from the operational layer, restructures this data for analysis across time and domains - weekly sell-through trends, quarterly margin shifts, promotion effectiveness - optimised for aggregation, historical comparison, and management reporting. Together, these two layers ensure that the same governed business truth drives both execution and intelligence.

2.1.1 Data Classification Framework

All data within the F&L ecosystem is classified along three dimensions. This classification determines what can be automated, what needs human curation, and what requires transformation to become machine-readable.

#	Classification	Description	Examples
1	Human-Created	Data generated by people through manual inputs, decisions, and judgement	Assortment plans, design briefs, vendor negotiations, markdown decisions, category strategy notes
2	Machine-Created	Data generated automatically by systems and automation	POS transactions, RFID scans, GRN logs, stock movement events, barcode scans, social media trend signals
3	Machine-Readable	Data that is structured, standardised, and directly consumable by AI/analytics systems	SKU attribute masters, hierarchy masters, price architecture files, BOM data, vendor scorecards

Key Principle: "Not all data is machine data - we have to classify it right." The data catalog must tag every data entity with its classification so that downstream systems know what they can consume directly versus what requires human input.

2.1.2 Data Architecture Layers

The data catalog is structured in layers, from most stable to most volatile. This layering is critical because it determines refresh cadences, caching strategies, and how intelligence systems interact with each data tier.

#	Category	Layer	Description	Update Frequency	Examples
1	Master	Fixed Masters	Core reference and static master data that forms the backbone of all operations	Rarely changes / Annually or less	Article Master, Store Master, Vendor Master, Hierarchy Master, EPD, Category Master, Location Master, Site Master, Format Definitions
2	Master	Periodic Masters	Masters that are refreshed on a regular cycle	Quarterly / Seasonal	Cluster Profiles, Season Calendar, Price Architecture, Size Ratios
3	Transactional	Daily / Transactional Data	Operational data flowing in continuously	Real-time / Daily	POS Sales, Stock Movements, GRN, RFID Scans, Transfer Events, Dispatch Records

Directive: All data lanes across sourcing systems must converge - every lane must be aligned so that any view, dashboard, or agent query can be built on top of a single, consistent data catalog.

2.1.3 Data Source Summary

67 activities across 6 lifecycle domains (Merchandising, Customer, Supplier, Asset, Finance, HR), powered by 120 formal SOPs, 35 data sources, and 45+ systems of record spanning Impetus, SAP, Databricks, and partner platforms across the full F&L value chain.

With the data classified and layered, the table below maps it to the **end-to-end F&L value chain** - spanning merchandising, customer, supplier, asset, finance, and HR lifecycles. Each row shows which data sources power the activity, what data entities are consumed, who is accountable, and how performance is measured.

2.1.4 Merchandising Lifecycle Management

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
1	Category Strategy	Category Strategy	- Impetus - POS - SAP F&L	OTB	Human-Created	R&M	Revenue & GM vs plan
2				Assortment Plan	Machine-Created		
3				Season Calendar	Machine-Readable		
4				Social Signals	Machine-Created		
5				Search Trends	Machine-Created		
6				Bestseller Ranks	Machine-Created		
7				POS Transactions	Machine-Created		
8				Sell-Through %	Machine-Created		

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
9				ROS	Machine-Created		
10	Pricing Strategy	Retail Price Setting	- Impetus - POS - SAP F&L	Cost Price	Machine-Readable	R&M	GM %, Price index vs competition
11				MRP	Machine-Readable		
12				Intake Margin	Machine-Created		
13				Landed Cost	Machine-Created		
14				Price Architecture	Machine-Readable		
15				POS Transactions	Machine-Created		
16				Revenue by Channel/Store	Machine-Created		
17	Demand Planning	- Demand Forecasting - Buy Planning & OTB Management	- Impetus - POS - SAP F&L	OTB	Human-Created	Channel Planning	Forecast accuracy, Stock availability %
18				Assortment Plan	Machine-Created		
19				Size Ratios	Machine-Readable		
20				Cluster Profiles	Machine-Readable		
21				POS Transactions	Machine-Created		
22				Sell-Through %	Machine-Created		
23				Stock Positions	Machine-Created		
24				RFID EPC	Machine-Created		
25				Ageing	Machine-Created		

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
26	Assortment Planning	- Master Assortment & Range Planning - Product Design & Trend Intelligence - Store Cluster Classification & Management - Store-level Assortment Planning - Assortment / SKU Performance Monitoring	- Impetus - Catalog Engine	OTB	Human-Created	R&M	SPSF, GMROF
27				Assortment Plan	Machine-Created		
28				Options	Machine-Readable		
29				SKUs	Machine-Readable		
30				Style Codes	Machine-Readable		
31				Size-Colour Matrix	Machine-Readable		
32				Tech-Pack	Machine-Created		
33				BOM	Machine-Readable		
34				Social Signals	Machine-Created		
35				Competitor Data	Machine-Created		
36				SKU Attributes	Machine-Readable		
37				Category Hierarchy	Machine-Readable		
38	Space Planning (including planogramming)	- Space Planning & Planogramming - Planogram Execution & Compliance Tracking	- SAP F&L - Impetus	Hierarchy Master	Machine-Readable	Channel Planning	GMROF
39				Site Master	Machine-Readable		
40				Category Master	Machine-Readable		
41				Location Master	Machine-Readable		
42				Stock Positions	Machine-Created		
43				Floor vs. Backroom Split	Machine-Created		
44	Order Confirmation	- Purchase Requisition Creation & Approval - Purchase Order Management	Impetus	PO	Human-Created	R&M	PR to PO cycle time
45				Vendor Master	Machine-Readable		
46				Vendor Score	Human-Created		

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
47				OTIF	Machine-Created		
48				Compliance Docs	Human-Created		
49				Cost Price	Machine-Readable		
50				MRP	Machine-Readable		
51				Intake Margin	Machine-Created		
52	Logistics & Warehousing	- Transport Planning - Execution - Dispatch - Warehousing - Dark Store Operations	- Anytrack - Impetus - SAP F&L	Carton ID	Machine-Created	Supply Chain	SLA adherence (TAT), Supply Chain cost per unit
53				Barcode	Machine-Created		
54				ASN	Machine-Created		
55				Shipment Manifest	Machine-Created		
56				Production Order	Human-Created		
57				Line Output	Machine-Created		
58				Fabric Specs	Human-Created		
59				Transfer Events	Machine-Created		
60				Dispatch Records	Machine-Created		
61				GRN	Machine-Created		
62				Put-Away Location	Machine-Created		
63	Inventory & Replenishment Planning	- Inventory Norming - Inventory Performance Tracking & RFID Management - Allocation & Replenishment	- SAP F&L - Impetus	Stock Positions	Machine-Created	Channel Planning	Days On Hand
64				RFID EPC	Machine-Created		
65				Ageing	Machine-Created		
66				Floor vs. Backroom Split	Machine-Created		

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
67				OTB	Human-Created		
68				Vendor Invoice	Machine-Created		
69				GRN Match	Machine-Created		
70				PO vs. Invoice Variance	Machine-Created		
71	IST, RTW & RTV Execution	- Return Logistics Tracking - Inter-store Transfer Tracking	- SAP F&L - Impetus	Transfer Events	Machine-Created	Channel Planning	IST, RTW, RTV SLA adherence
72				Dispatch Records	Machine-Created		
73				Receipt Logs	Machine-Created		
74				Vendor Invoice	Machine-Created		
75				GRN Match	Machine-Created		
76				Debit/Credit Notes	Machine-Created		
77	Markdown Management & Liquidation (inc. IST, RTW, RTV)	- Markdown Planning - Inventory Liquidation Management - Returns Planning - Inter-store Transfer Planning	- POS - SAP F&L - Impetus	POS Transactions	Machine-Created	R&M	Delivered GM %, Aged inventory %
78				Sell-Through %	Machine-Created		
79				ROS	Machine-Created		
80				Stock Positions	Machine-Created		
81				Ageing	Machine-Created		
82				Cost Price	Machine-Readable		
83				MRP	Machine-Readable		
84				Intake Margin	Machine-Created		
85	VM Planning	VM and In-store Marketing Planning	- Impetus - Catalog Engine	Options	Machine-Readable	R&M	Footfall and conversion
86				SKUs	Machine-Readable		

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
87				Style Codes	Machine-Readable		
88				Tech-Pack	Machine-Created		
89				SKU Attributes	Machine-Readable		
90				Product Images	Machine-Readable		
91	Promotion Planning	- Promotion Design - Pricing & Promotion Execution	- POS - SAP F&L - Impetus	POS Transactions	Machine-Created	R&M	Delivered GM %, Promo ROI
92				Revenue by Channel/ Store	Machine-Created		
93				Cost Price	Machine-Readable		
94				MRP	Machine-Readable		
95				Price Architecture	Machine-Readable		
96	Online order fulfilment & returns	- Omni-channel Fulfilment - Order SLA Monitoring	- SAP F&L - POS	Stock Positions	Machine-Created	Channel Planning / Supply Chain	Omni-channel order SLA adherence
97				RFID EPC	Machine-Created		
98				POS Transactions	Machine-Created		
99				GRN	Machine-Created		
100				Inbound Quantity	Machine-Created		
101				Quality Check Results	Machine-Created		

2.1.5 Customer Lifecycle Management

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
1	In-store customer journey curation	- In-Store Customer Experience Management	- POS	Footfall counters	Machine-Created	Operations	Conversion %, Customer NPS
2		- In-Store Self-Checkout (Scan & Go)	- PCM - People Counting Module				
3		- In-Store Endless Aisle and Digital Ordering	- Anytrack				
4		- Store Opening Operations	- SAP F&L				
5		- Store Closing Operations	- Impetus				
6		- Inbound Inventory Management	- LMS (Opigno)				
7		- In-Store Inventory Management					
8	Online platform customer journey curation	- In-Store Inventory Management		POS conversion	Machine-Created		
9		- In-Store Inventory Control		VM checklist completion	Human-Created		
10		- Cash Handling & Reconciliation		Staff readiness	Human-Created		
11		- NHQ to Store Communications		Stock Positions	Machine-Created		
		- VM Execution & Compliance Tracking		RFID EPC	Machine-Created		
		- Store Escalation Management		Cash reconciliation	Machine-Created		
		- Workforce Scheduling & Rostering					
		- Incentive Management					
		- Store Workforce Training & Capability Development					
8	Online platform customer journey curation	- Product Listing Creation & Activation	- Catalog Engine	SKU Attributes	Machine-Readable	Online Ops	Conversion %, Customer NPS
9		- Content QA & Product Detail Page Approval	- SF CMS				
10			- Google Analytics (BigQuery)				
11							
				Product Images	Machine-Readable		
				Site content	Human-Created		
				Clickstream	Machine-Created		

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
12				PDP views	Machine-Created		
13	Post purchase experience (incl. query / complaint resolution)	- Post-purchase Service Fulfilment - Customer Query Intake - Issue Resolution Management - Escalation Management - Customer Query Resolution (Digital Service Channels)	- Haptik - Genesys - Hybris - OMS - Order Management System	Chat transcripts	Machine-Created	Customer Service	CSAT, First Contact Resolution %
14				Call center logs & reason codes	Machine-Created		
15				Tickets	Machine-Created		
16				Order status	Machine-Created		
17				NPS surveys	Human-Created		
18	Returns / refunds execution	- Refund & Replacement Execution - Returns Reconciliation	- OMS - Order Management System - WMS - Warehouse Management System - SAP F&L	Return initiated/ received	Machine-Created	Operations	Customer NPS, TAT for returns
19				Graded/ routed (restock/ transfer/B2B/ export)	Machine-Created		
20				Refund status	Machine-Created		
21				Debit/Credit Notes	Machine-Created		

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
22	Customer acquisition (Branding / Events / Large campaigns)	- Customer Insights Generation - Brand Strategy and Communication Planning - Marketing Budget ROI and Governance - Customer Segmentation & Cohort Design - TG Definition & Campaign Planning - Channel Mix Optimization - Campaign Performance Measurement - Traditional & Digital Marketing Execution - Creative Strategy and Content Governance	- Ad Platforms - Locobuzz - Galleri5 - Clevertap - Google/ Meta - Google Analytics (BigQuery)	Campaign details	Human-Created	Marketing	Footfall / Traffic, CAC, Marketing ROI, Influencer ROI
23				Customer interactions	Machine-Created		
24				Trend mentions	Machine-Created		
25				Hashtag velocity	Machine-Created		
26				Creator engagement	Machine-Created		
27				Influencer details	Human-Created		
28				Impressions	Machine-Created		
29				Engagement	Machine-Created		
30	Personalization / Next-Best-Offer Engine (Omni-channel)	- Customer Journey Mapping & Personalization - Offer Recommendation Engine Management - Affordability Solutions Management	- UCP - SF database - Google Analytics (BigQuery) - Amplitude - Wallet	Customer profile	Machine-Readable	Marketing	Conversion %, CLTV, CRM Activation Rate
31				Identity resolution (online/offline)	Machine-Created		
32				Preferences	Human-Created		
33				Segments	Machine-Created		
34				Affinity data	Machine-Created		
35				Cohorts	Machine-Created		
36				Loyalty ledger	Machine-Created		

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
37	Customer retention	- Customer Lifecycle & Churn Management - NPS CSAT and VoC (Voice of Customer) Capturing	- UCP - Clevertap - Wallet	NPS surveys	Human-Created	Marketing	Retention Rate, CLTV
38				Post-purchase CSAT	Human-Created		
39				Points/benefits earned & redeemed	Machine-Created		
40				Customer segments	Machine-Created		
41				Churn indicators	Machine-Created		

2.1.6 Supplier Lifecycle Management

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
1	Supplier strategy & governance	Supplier Identification & Evaluation	- SAP F&L - Impetus	Supplier master	Machine-Readable	Sourcing POD	Strategic supplier mix %, Supplier performance score
2				Supplier onboarding/changes	Human-Created		
3				Vendor Score	Human-Created		
4	Trade terms, negotiations and Joint Business Plan (JBP)	- RFX Management (RFI/RFQ/RFP) - Contract & TOT Negotiation & Finalization - Rebate Management	- SAP F&L - Impetus	Cost per option	Human-Created	Sourcing POD	Intake gross margin %, Rebate realization %
5				Vendor Master	Machine-Readable		
6				Contract terms	Human-Created		
7				Rebate schedules	Human-Created		
8	Supplier onboarding & relationship management (incl. dispute resolution)	- Supplier Onboarding - Supplier Issue & Escalation Handling	- SAP F&L - Impetus	Supplier master	Machine-Readable	Sourcing POD	Supplier NPS, Supplier onboarding TAT
9				Onboarding status	Machine-Created		
10				Compliance docs	Human-Created		

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
11				Escalation logs	Machine-Created		
12	Marketplace Seller experience management	Marketplace Seller Experience Management	Seller Central	Seller onboarding	Human-Created	Marketplace Management	Marketplace seller NPS
13				Catalog management	Human-Created		
14				Fulfilment status	Machine-Created		
15	Ordering, goods receipt and payment (PO to Pay)	- Goods Receipt & Service Confirmation - Invoice Processing & Reconciliation - Vendor Payment Execution	- SAP F&L - Impetus	PO create/change	Human-Created	Sourcing POD	Order cycle time, Invoice cycle time
16				ASN	Machine-Created		
17				GRN	Machine-Created		
18				Vendor Invoice	Machine-Created		
19				Accounts Payable	Machine-Created		
20				Payment status	Machine-Created		
21	Quality assurance	- Quality Standards & Norm Setting - Quality Audits & Execution	Impetus	Defect logs	Machine-Created	Quality	Quality pass rate %, Quality audit score
22				Fit feedback	Human-Created		
23				Alteration rates	Machine-Created		
24				Inspection and QC Master	Machine-Readable		
25				Quality pass rate	Machine-Created		
26	Supplier performance management	Supplier Performance Management	- SAP F&L - Impetus	OTIF	Machine-Created	Sourcing POD	OTIF %, Supplier scorecard rating
27				Leadtime Variance	Machine-Created		
28				Defect rates	Machine-Created		
29				Cost variance	Machine-Created		
30				Supplier scorecard	Machine-Readable		

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
31	Innovation / New Product Development (NPD) collaboration	PL Concept Design & Development (Private Label)	Impetus	Design brief	Human-Created	Research & Development	NPD hit rate %, Innovation revenue contribution %
32				Sample iterations	Human-Created		
33				Tech-Pack	Machine-Created		
34				BOM	Machine-Readable		
35				Social Signals	Machine-Created		
36				Competitor information	Machine-Created		

2.1.7 Asset Lifecycle Management

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
1	Site identification & finalization (Network Planning)	- Competitor Mapping & Catchment Study - Network Footprint Design - Capacity & Throughput Planning - Scenario Modelling & Network Phasing - Network Performance Monitoring - WH Design Finalization - Dark Store Design Finalization - Market & Site Scanning - Lease Negotiation - Legal & Compliance Clearance	- Board - SAP F&L	Workflow	Human-Created	Network Planning / Supply Chain / Real Estate	Network Utilization, Network Cost-to-Serve, Lease cost per sq ft, Design Cycle Time
2				Capex	Human-Created		
3				Approvals	Human-Created		
4				Catchment data	Machine-Created		
5				Network utilization	Machine-Created		
6				Lease cost per sq ft	Human-Created		
7				Throughput feasibility	Machine-Created		
8	Capex approval management	Feasibility & Business Case Assessment	- Board - SAP F&L	Capex budget	Human-Created	F&L Asset	Approval cycle time, Payback period
9				Business case	Human-Created		

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
10				Payback projections	Human-Created		
11				Approval status	Machine-Created		
12	New Asset Opening	- Store Design Finalisation - Utility & Infrastructure Assessment - Site Handover - Fit-Out Execution - Pre-Opening Readiness & Compliance	- Board - SAP F&L	NSO workflow	Human-Created	F&L Asset / EPC	NSO plan adherence %, Budget adherence %
13				Capex	Human-Created		
14				Approvals	Human-Created		
15				Design specs	Human-Created		
16				Compliance status	Machine-Created		
17	Asset Maintenance & Upkeep	- Maintenance & Facilities Ticketing - Asset Health Monitoring - IoT & Automation Setup - Store Hygiene Management - Operational Audits & Compliance - Scrap Management - Store Performance Monitoring	- Facility Management - SAP F&L - Impetus - PCM - People Counting Module	Asset management	Human-Created	Operations	Store uptime %, Maintenance TAT
18				Equipment	Machine-Readable		
19				Work orders	Machine-Created		
20				VM checklist completion	Human-Created		
21				Footfall counters	Machine-Created		
22				Store uptime	Machine-Created		
23	Asset Refit, Refresh & De-Hiring	- Store Remodelling & Refresh - Store Decommissioning - Asset Receiving & Returns	- Board - SAP F&L	Refit budget	Human-Created	F&L Asset / EPC / Real Estate	Refit ROI / sales uplift, Capex adherence %, Downtime duration, Asset recovery %, Dehire TAT
24				Capex adherence	Machine-Created		
25				Downtime duration	Machine-Created		
26				Asset recovery	Machine-Created		

#	Activity	SOPs	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability	KPI
27	Security & Loss Prevention	- Store Safety Management - 3PL Guard Management - Incident Review & Investigation - Access Control Management - Store Safety Audit - High-Value SKU Monitoring - Security Device Health Monitoring - SLP Training & Readiness - Emergency Response Management - Shrinkage Monitoring	- SAP F&L - LMS (Opigno)	Dehire TAT	Machine-Created		
28				Closure cost vs plan	Human-Created		
29				Shrinkage data	Machine-Created	HSE / Security & Loss Prevention	Shrinkage %, Safety incident rate
30				Safety incident logs	Machine-Created		
31				Access control logs	Machine-Created		
32				Security device health	Machine-Created		
33				Dump Qty	Machine-Created		
34				Losses Qty	Machine-Created		
35				Losses Value	Machine-Created		

2.1.8 Finance

#	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability
1	Pricing Engine	Price changes	Machine-Created	Finance
2	POS	Promotions - offline	Machine-Created	Finance
3	Promotion Engine	Promotions - online	Machine-Created	Finance
4	POS	Coupon Usage	Machine-Created	Finance
5	POS + OMS	Line-item transactions	Machine-Created	Finance
6		Discounts	Machine-Created	
7		Returns	Machine-Created	
8		Taxes	Machine-Created	
9	SAP F&L	COGS	Machine-Created	Finance
10		Logistics	Machine-Created	
11		Returns cost	Machine-Created	
12		Markdown cost	Machine-Created	
13		Shrink	Machine-Created	
14	SAP F&L	General Ledger Accounts	Machine-Readable	Finance
15		Balances	Machine-Readable	
16		Group Company Transactions	Machine-Readable	

#	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability
17	SAP F&L	Accounts Payable	Machine-Created	Finance
18		Vendor Balances	Machine-Created	
19		Open Items	Machine-Created	
20	SAP F&L	Margins	Machine-Created	Finance
21		Operations Cost	Machine-Created	
22		EBITDA	Machine-Created	
23	SAP F&L	Receivables	Machine-Created	Finance
24		Payables	Machine-Created	
25		Inventory	Machine-Created	
26	SAP F&L	GST	Machine-Created	Finance
27		TDS	Machine-Created	
28		Reco Balances	Machine-Created	

2.1.9 HR

#	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability
1	SAP - Retail HR	Staffing changes	Human-Created	HR
2		Role mapping	Human-Created	
3	Zing HR	Skill tag	Human-Created	HR
4	SAP - Retail HR	Incentive rules	Human-Created	HR
5	SAP - Retail HR	Payouts	Machine-Created	HR
6		KPI attainment	Machine-Created	
7	SAP - Retail HR	Insurance	Human-Created	HR
8		Policies	Human-Created	
9		IT Assets	Machine-Readable	
10	SAP - Retail HR	Onboarding	Human-Created	HR
11		Progression	Human-Created	
12		Transfer	Human-Created	
13	SAP - Retail HR	Personal	Human-Created	HR
14		Education	Human-Created	
15		Experience	Human-Created	
16		Nominations	Human-Created	
17	SAP - Retail HR	Leaves	Machine-Created	HR
18		Attendance	Machine-Created	
19	SAP - Retail HR	Attrition	Machine-Created	HR
20		Settlement	Machine-Created	
21		Exit Interview	Human-Created	
22	Opigno LMS	Course	Machine-Readable	HR

#	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability
23		Learning Hours	Machine-Created	
24		Completion	Machine-Created	
25		Feedback	Human-Created	
26	SAP - Retail HR	Goal Setting	Human-Created	HR
27		PMS	Human-Created	
28		Store Excellence	Human-Created	
29	SAP - Retail HR	Payouts	Machine-Created	HR
30		Reimbursements	Machine-Created	
31	SAP - Retail HR	Travel Requests	Human-Created	HR
32		Status	Machine-Created	
33	SAP - Retail HR	Job	Machine-Readable	HR
34		Positions	Machine-Readable	
35		Required	Human-Created	
36		Available	Machine-Created	
37		Reporting Matrix	Machine-Readable	
38	SAP - Retail HR	Candidate	Human-Created	HR
39		Interview	Human-Created	
40		Talent Marketplace	Human-Created	
41	SAP - Retail HR	Queries	Machine-Created	HR
42		Area	Machine-Readable	
43		Closure	Machine-Created	
44	SAP - Retail HR	Category	Machine-Readable	HR
45		Resolution Status	Machine-Created	
46	Survey Systems	Corporate Survey	Human-Created	HR
47		Employee Survey	Human-Created	
48		Feedbacks	Human-Created	
49		Employee Relations	Human-Created	
50	SAP - Retail HR	Recognitions	Human-Created	HR
51		Rewards	Human-Created	
52		Nominations	Human-Created	
53	Facility Management	Asset management	Machine-Created	HR
54		Equipment	Machine-Readable	
55		Work orders	Machine-Created	
56		Task List	Machine-Readable	
57	HMS	PME Due	Machine-Created	HR
58		Completion	Machine-Created	
59		Policies	Human-Created	

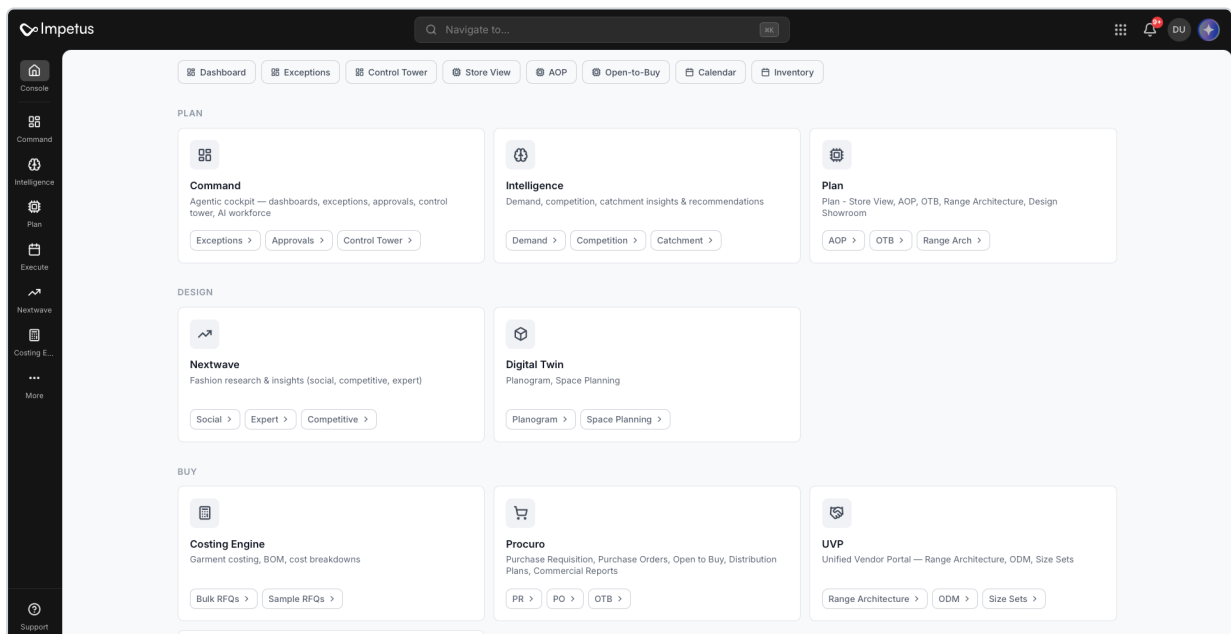
#	Data Sources	Key Data Entities / KPIs	Data Classification	Accountability
60	Betterplace	Headcount	Machine-Created	HR
61		Attendance	Machine-Created	
62		Profile	Human-Created	

3. Command Centre

3.1 Command Centre Architecture

Note: The Command Centre platform is currently under verification and is scheduled to launch by April 30, 2026.

The Impetus Command Centre is a Databricks-powered real-time intelligence layer that consumes data from 35 data sources, monitors 67 activities and 120 SOPs across 6 lifecycle domains against their KPIs, enforces guardrails, and surfaces exceptions that need human attention. Every alert, dashboard, and recommendation traces back to a specific workflow activity and data source. It is built on four pillars.

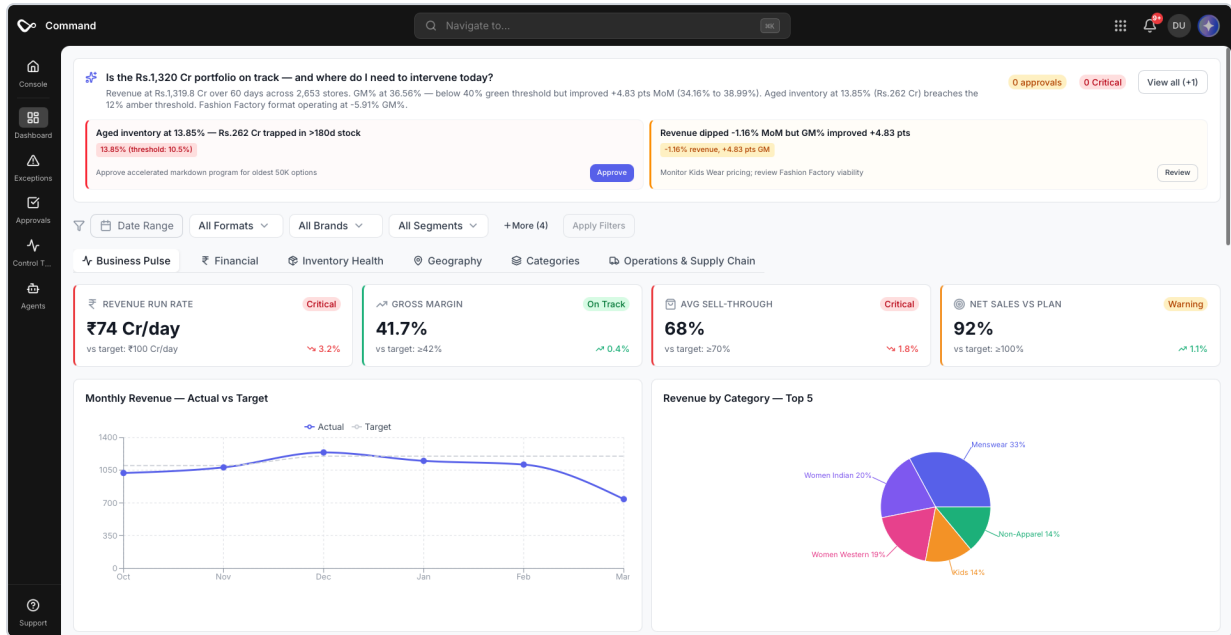


3.1.1 Dashboards

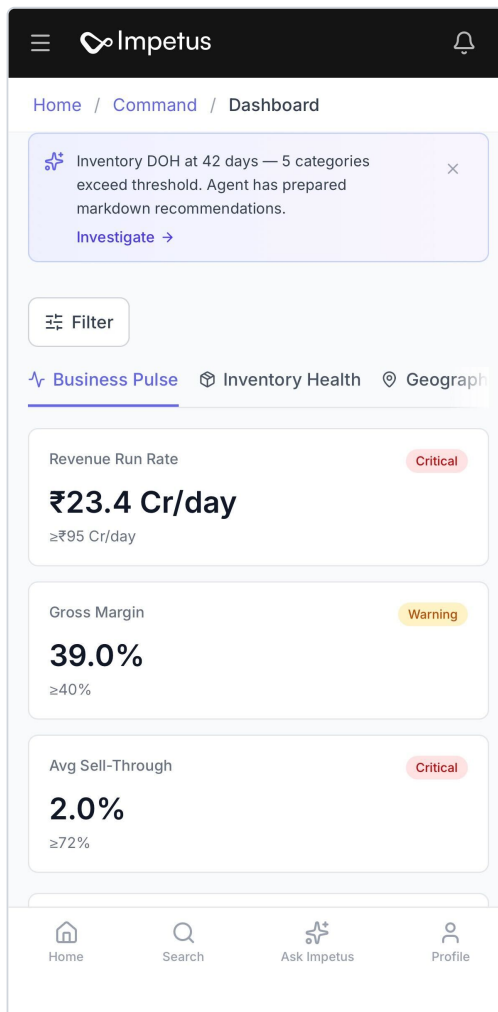
Real-time operational visibility across all 120 SOPs. Three role-based views:

- **CEO Dashboard** - KPI summary across all SOPs, portfolio-level revenue and margin tracking
- **R&M Head Dashboard** - Trend signals, sell-through, ageing, OTB status
- **Store Manager Dashboard** - Inventory positions, RFID compliance, display compliance

The dashboard surfaces four key metrics in real time: Revenue Run Rate, Gross Margin, Average Sell-Through Rate, and Net Sales vs Plan - each with on-track/off-track status and trend indicators.



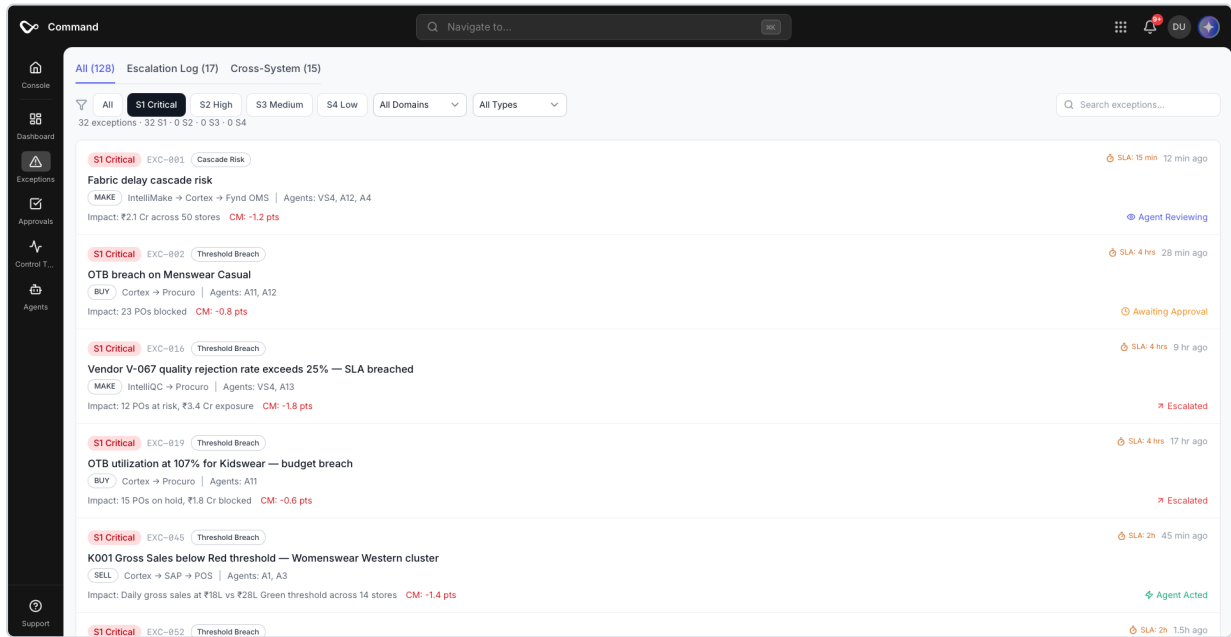
The same dashboard is accessible on mobile, with agent-generated alerts surfaced inline - for example, flagging inventory DOH thresholds and prepared markdown recommendations for immediate action.



3.1.2 Exception Management

System-triggered alerts with prescribed escalation paths. Each exception carries a severity level (Critical, High, Medium, Low), an auto-assigned owner, and an SLA for resolution. Examples:

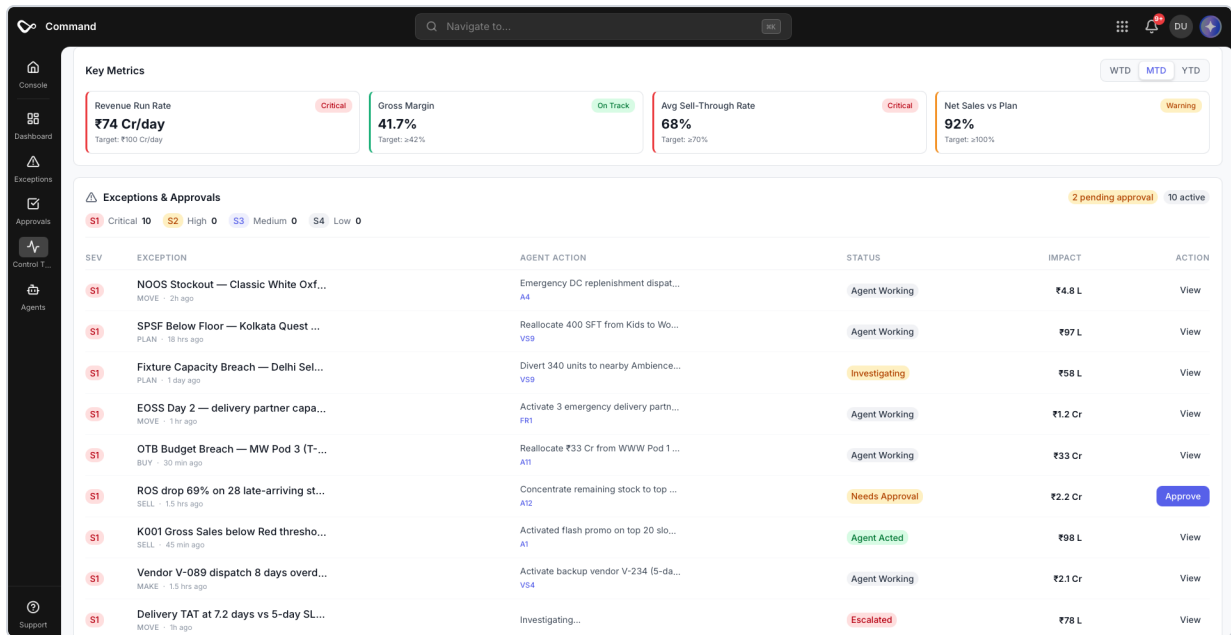
- **NOOS stockout** - 24hr resolution SLA
- **Fashion ageing >45 days unactioned** - escalation to R&M Head
- **Intake margin breach (<50%)** - immediate flag to R&M
- **Vendor delivery delay >3 days** - escalation to Sourcing
- **RFID accuracy <99%** - store-level flag



3.1.3 Approvals & Controls

Governance workflow with RACI-embedded approval chains. Every approval traces back to the SOP it enforces. Examples:

- **PO approval** - dual sign-off: Sourcing Head + R&M Head
- **Markdown triggers** - R&M Planner initiates, R&M Head approves same-day
- **OTB budget ceiling** - system-enforced, exception requires escalation



3.1.4 Control Tower

Cross-functional process orchestration with SLA tracking across the end-to-end value chain.

- **End-to-end delivery tracking** - vendor to DC to store, with TAT at each node
- **Omnichannel fulfilment SLA** - Metro same-day dispatch enforcement
- **Weekly cadence enforcement** - Trend Review (Monday), Inventory Review (Friday)

3.2 Explainability Framework

Every automated decision in the system carries an explainability trace - linking the recommendation back to its data source, the KPI it impacts, and the threshold that triggered it. Examples:

3.2.1 Markdown Triggers

System shows current sell-through %, days in store, projected sell-through at current ROS, and the specific threshold breached. Example: "Week 4 sell-through 22% < 25% threshold - Stage 1 markdown recommended." The trace links back to Sales and Stock Inventory data sources.

3.2.2 Replenishment Decisions

MSL (Minimum Stock Level) formula visible: (Weekly ROS x Lead Time) + Safety Stock. The system shows historical ROS trend and current inventory position alongside the recommendation. Powered by Stock Inventory and Merchandise Planning data.

3.2.3 Trend Signals

Each signal tagged with source (Ajoio search data / social scrape / POS fast-mover / competitor monitoring), confidence score, and 4-week conversion history. Signals feed into Category Strategy and Assortment Planning workflows.

4. Details of AI Agents

A mapping of the AI agents powering L4 (Agentic) workflows across Plan, Buy, Move, and Sell for F&L.

4.1 Preamble

The 90-day autonomy heat map (Paper 1, Section 1.5) identifies five workflows that reach L4 - where the system makes cross-domain decisions and the human sets intent rather than instructions. Each of these workflows is powered by a specific AI agent already in development or deployment on the Impetus platform. These agents operate on the same data catalog described in Paper 2 - the 67 activities, 120 SOPs, and 35 data sources that power the retail value chain. This paper provides a brief summary of each agent and maps it to the value chain track it operates in. It is not an in-depth write-up on agent internals - detailed architecture and implementation specifications are maintained separately.

4.2 Plan

Planning Agent (Impetus Cortex) powers Space Planning at L4. It builds a store-realistic Plan of Record and converts it into fixture-level and slot-level VM layouts. When conditions change - a delayed shipment, an underperforming category, a competitive signal - the agent generates an exception-led replan. The planner reviews the replan, not the plan. The agent does not replace planning. It makes planning continuous.

Store Intelligence Agent operates as the spatial intelligence backbone across all Plan workflows. It enables geo-led expansion, catchment-aware routing, store-level personalisation, and SLA optimisation. Vista provides the demand and competitive context that Cortex uses to build plans.

4.3 Buy

Trend-to-Design Agent powers the design-to-buy pipeline that feeds Order Placement at L4. It transforms trend signals and brand context into manufacturable design options and auto-generates complete tech packs - specifications, BOM, trims, artwork - ready for vendor handoff. The buyer reviews a finished package, not a brief. The agent compresses what was a multi-month cycle into days. The Impetus platform connects this to **Costing Engine** and **Purchase Automation** which help execute the buying decision once the agent has prepared it.

4.4 Move

Move workflows target L3 (Intelligent) across the board - predictive, continuous, demand-signal driven. The agent layer here is the **Manufacturing Execution System (IntelliMake)**: a unified AI-native vendor OS that connects every stage of the factory. Pre-production order summaries, daily production

reporting, quality audit modules, capacity declarations, and AI orchestration replace the email-and-phone-call visibility that most supply chains run on. **Vendor Management at L4** is powered by the same supplier graph: real-time capacity tracking, quality scoring, compliance monitoring, and automated reconciliation. The system manages the relationship operationally. The acquisition strategist manages it strategically.

The Command Centre's **Agentic Control Tower** governs the Move track. It detects exceptions (a delayed shipment, a quality breach, a capacity constraint), quantifies the impact in rupees, identifies the mitigation, and either executes it or surfaces it for approval. The control tower handles the exceptions. The supply chain team handles the disruptions that require human judgement.

4.5 Sell

In-Store Experience Agent powers Store Operations and Visual Merchandising at L4. It fuses CCTV and store data (POS, NPS, inventory) to track footfall, dwell time, staff gaps, and service quality - turning signals into real-time actions via a continuous Sense, Analyse, Act, Verify loop. The store manager receives a daily brief, not a weekly review deck. The agent does not replace the store team. It makes every store a market of one.

Impetus Store Ops App is the execution surface for the In-Store Experience Agent - task management, planogram compliance, inventory scanning, KPI dashboards, staff scheduling, and an AI-native assistant, all on a single mobile interface.

Agentic Marketing (via UCP) operates across the Sell track for campaigns that reach L3-L4. It autonomously builds and optimises campaigns - audiences, creatives, timing, and measurement - using customer signals rather than calendar briefs. The marketing team defines brand rules and reviews output. The agent does the rest.

AI Cataloguing and Photoshoots compress content creation from days to hours. A mannequin image becomes a model-on photoshoot, an enriched catalog page, and a generated video - scalable, consistent, and fast. **Generative Media** extends this to product videos, video ads, and social content at a cadence no human team can sustain.

More agents for retail workflows across the value chain are in development and will be detailed in subsequent updates.